CRANE AI-Cyber SIG — Presentation Preparation

SABRE: An Uncertainty-Aware Modular Semantic–Behavioural Triage Framework for Adaptive Ransomware Detection

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1 · Opening Statement

Deliver this before advancing past the title slide (≈20 seconds):

> *"Every ransomware detector in production today gives you one of two answers: threat or not. But when that answer is wrong — and it will be — you have no way of knowing the system was uncertain. SABRE asks a different question: not just what is this process doing, but how confident should we be in that assessment. And then it acts on that confidence."*

Then advance to the Agenda slide and say: "Here is what we will cover in the next 15 minutes."

2 · Slide Timing Guide (15 min total)

Slide	Title	Target
1	Title + opening statement	1 min
2	Agenda	30 sec
3	The Problem	1.5 min
4	SABRE Architecture	2 min
5	Triage Policy	2 min
6	Results	2 min
7	Explainability	2 min
8	SABRE-II	2 min
9	Conclusions	1 min
	Total	**14 min (1 min buffer)**

3 · Numbers to Know Cold (no looking at slides)

Metric	Value	Context
Semantic AUROC	**0.996**	RDset — 138K PE files
Behavioural AUROC	**0.963**	RanSMAP — random split
Family holdout AUROC	**0.706**	REvil + WannaCry unseen

Concurrent exec. detect	**1.5%**	AESCrypt + REvil mixed
ECE (calibration)	**0.094 → 0.018**	SABRE → SABRE-II
Balanced accuracy	**0.525 → 0.866**	Fixed → Bayesian TPE
Escalation rate	**85% → 25%**	Fixed → Bayesian TPE
Bayesian TPE trials	**300**	Policy optimisation
MC Dropout passes	**T = 10**	Per agent, per sample
Inference latency	**≈ 3.7 ms**	Full pipeline, NVIDIA GB10

4 · Anticipated Question: Monte Carlo Dropout

Q: What is MC Dropout and why did you use it?

*"Standard neural networks at inference time turn dropout off — you get a single deterministic output. Monte Carlo Dropout keeps dropout active at test time. We run the same input through the network T=10 times, each time randomly dropping different neurons. The **mean** of those T predictions gives us **p■** — the detection score. The **standard deviation** gives us σ — the uncertainty. It approximates Bayesian inference without redesigning the architecture. The key appeal: ten forward passes costs about 4× a single pass, and we get a principled uncertainty estimate in return."*

Q: Why T=10 and not more?

"T=10 is the standard from Gal & Ghahramani (2016) and gives a stable variance estimate at low cost. We verified empirically that σ stabilises by T=10 — going to T=50 moves the uncertainty estimates by less than 1%."

Q: Is the uncertainty actually calibrated?

"That is exactly what motivated the calibration work in SABRE-II. Raw MC Dropout uncertainty is not calibrated by default. We applied temperature scaling and measured Expected Calibration Error dropping from 0.094 to 0.018, meaning the uncertainty now reflects actual error rates. That is what makes it trustworthy enough to gate autonomous containment decisions."

Q: What does σ actually represent — is it epistemic or aleatoric?

"It is epistemic uncertainty — uncertainty due to limited or ambiguous training data, not irreducible noise in the signal. That distinction matters operationally: epistemic uncertainty decreases as the model sees more examples of a family. Aleatoric uncertainty would not. When a novel family arrives, σ is high because the model has not seen it — that is the correct signal to escalate to an analyst."

5 · Anticipated Question: Triage Thresholding

Q: How do you set the thresholds?

"We have four thresholds: τ and τ_high on the risk score p■, and κ and κ_low on the uncertainty σ. In the original SABRE they were hand-tuned — and the result was an 85% escalation rate, which is operationally useless. In SABRE-II we replaced that with Bayesian optimisation using a Tree-structured Parzen Estimator over 300 trials. The objective was: maximise balanced accuracy minus a penalty on escalation rate. That single change brought escalation from 85% down to 25%

while balanced accuracy went from 0.525 to 0.866."

Q: How do you control the cost trade-off?

"The objective is J(τ,κ) = balanced_accuracy – λ × escalation_rate. λ is the operational cost parameter. Higher λ penalises escalation more heavily — appropriate where analyst capacity is the bottleneck. Lower λ prioritises detection, accepting more escalations. In practice, λ is set based on the ratio of analyst-hour cost to the estimated cost of a missed detection — a deployment decision, not a modelling one."

Q: What happens when the threat landscape drifts — do you retrain?

"No, and that is a deliberate design choice. The triage policy is separated from the neural agents. If concept drift occurs — a new family emerges, threat patterns shift — you recalibrate the thresholds on a rolling buffer of analyst-labelled decisions without touching the agents at all. We discuss two concrete mechanisms in the limitations section: sliding-window recalibration, and Bayesian online updating of the threshold posterior."

6 · Other Likely Questions

Q: How does this compare to CrowdStrike / SentinelOne?

"Commercial EDR tools use confidence scores internally but do not expose uncertainty to the analyst — you get a verdict and a severity score, not a principled 'I don't know.' SABRE makes σ a first-class output that routes the decision, not just an internal weight. The analyst can see why a sample was escalated, not just that it was."

Q: Family holdout AUROC dropped to 0.706 — is that good enough?

"The honest answer is: for confident autonomous containment, no — and we say so. But that is precisely the point. When σ_beh rises on unseen families, the system routes to ESCALATE, not AUTO-CONTAIN. The uncertainty mechanism is doing its job: it knows when it does not know, and it passes the decision to a human. The 0.706 AUROC is on a held-out set with no training examples — the system degrades gracefully rather than silently."

Q: What datasets did you use?

"Three: RDset — 138,047 PE files for the semantic agent. RanSMAP — 35,857 ten-second telemetry windows from 15 hardware configurations across 6 ransomware families and 6 benign applications, captured under BitVisor hypervisor. And RansKnow — 307 ATT&CK threat-intelligence transcripts for the knowledge agent in SABRE+."

7 · Practice Checklist

- ■ Run a full timed rehearsal — stop at 15 min regardless of where you are
- ■ Practice the Architecture slide (slide 4) verbally without reading the figure
- ■ Practice the SABRE → SABRE-II transition on slide 8 as one fluid narrative
- ■ Recite all numbers in Section 3 from memory before the day
- ■ Record yourself once on your phone and watch it back — catches filler words
- ■ Prepare a one-sentence answer to 'what is your contribution over prior work?'

■ Know which slide has which figure — don't hunt for them during the talk

Tip: end every answer with a number. Audiences can tell the difference between someone who understands the work and someone who has memorised a description.